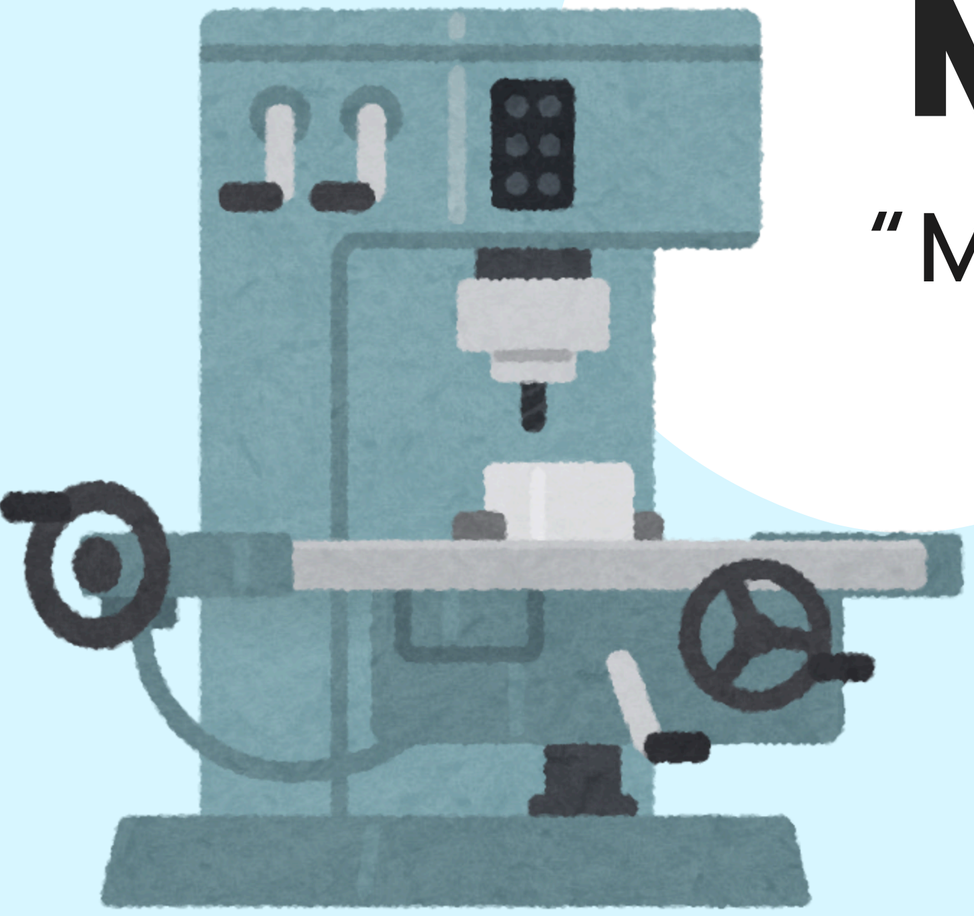
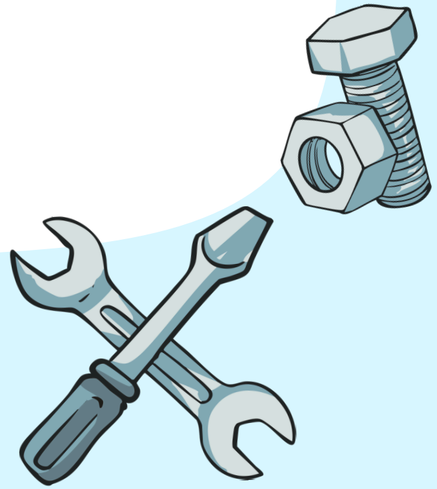
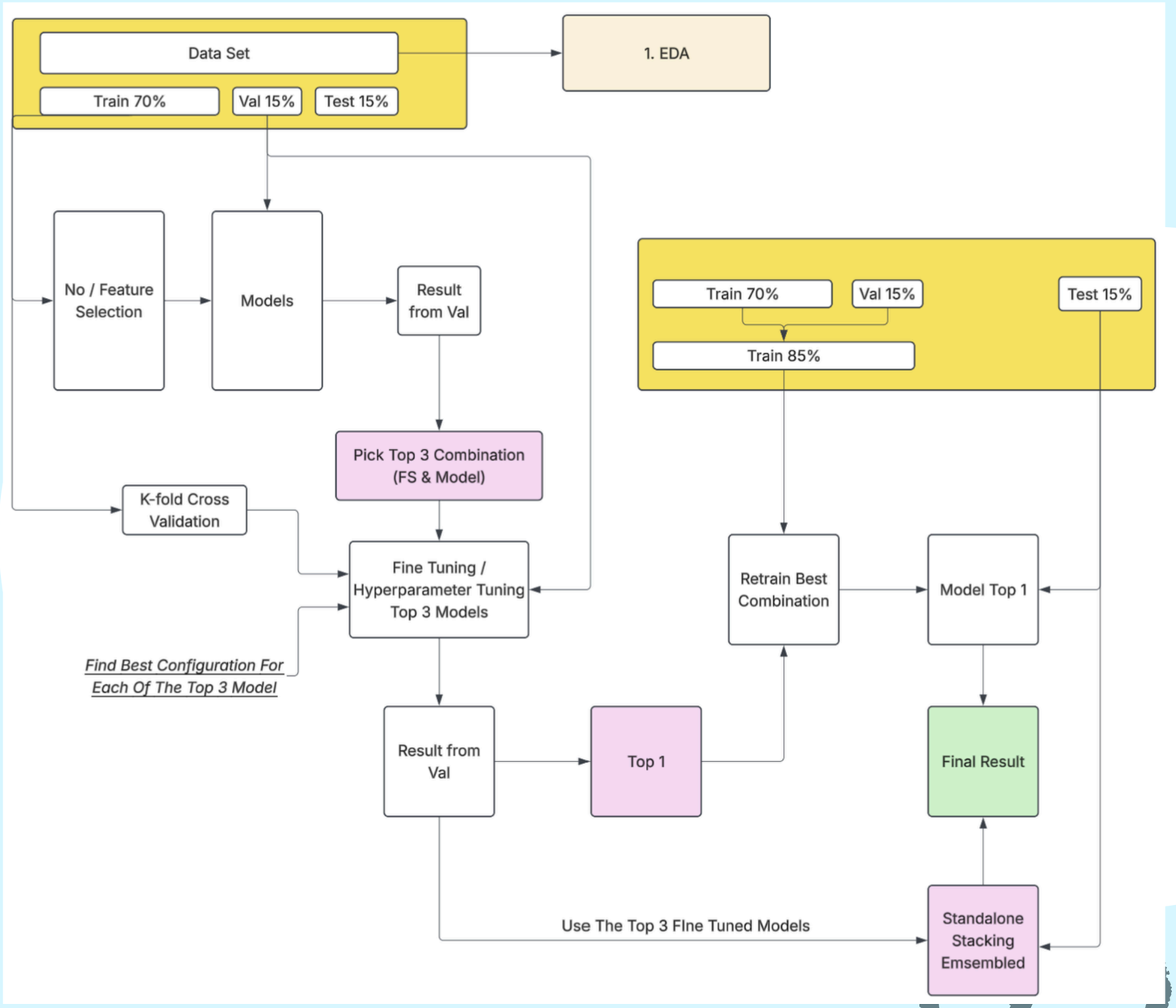


ML Term Project

"Machine Failure Classification"





Overview...

- Introduction
- Exploratory Data Analysis
- Pre-processing
- Modeling and Evaluation
- Model Selection and Fine-Tuning
- Final Model Evaluation
- Conclusion

Introduction

Introduction

STEPHAN MATZKA · UPDATED 4 YEARS AGO

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Predictive Maintenance Dataset (AI4I 2020)

The original dataset of a synthetic milling process for classification and XAI.



```
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   UDI                     10000 non-null  int64
 1   Product ID              10000 non-null  object
 2   Type                    10000 non-null  object
 3   Air temperature [K]     10000 non-null  float64
 4   Process temperature [K] 10000 non-null  float64
 5   Rotational speed [rpm]  10000 non-null  int64
 6   Torque [Nm]             10000 non-null  float64
 7   Tool wear [min]         10000 non-null  int64
 8   Machine failure         10000 non-null  int64
 9   TWF                     10000 non-null  int64
 10  HDF                     10000 non-null  int64
 11  PWF                     10000 non-null  int64
 12  OSF                     10000 non-null  int64
 13  RNF                     10000 non-null  int64
dtypes: float64(3), int64(9), object(2)
```

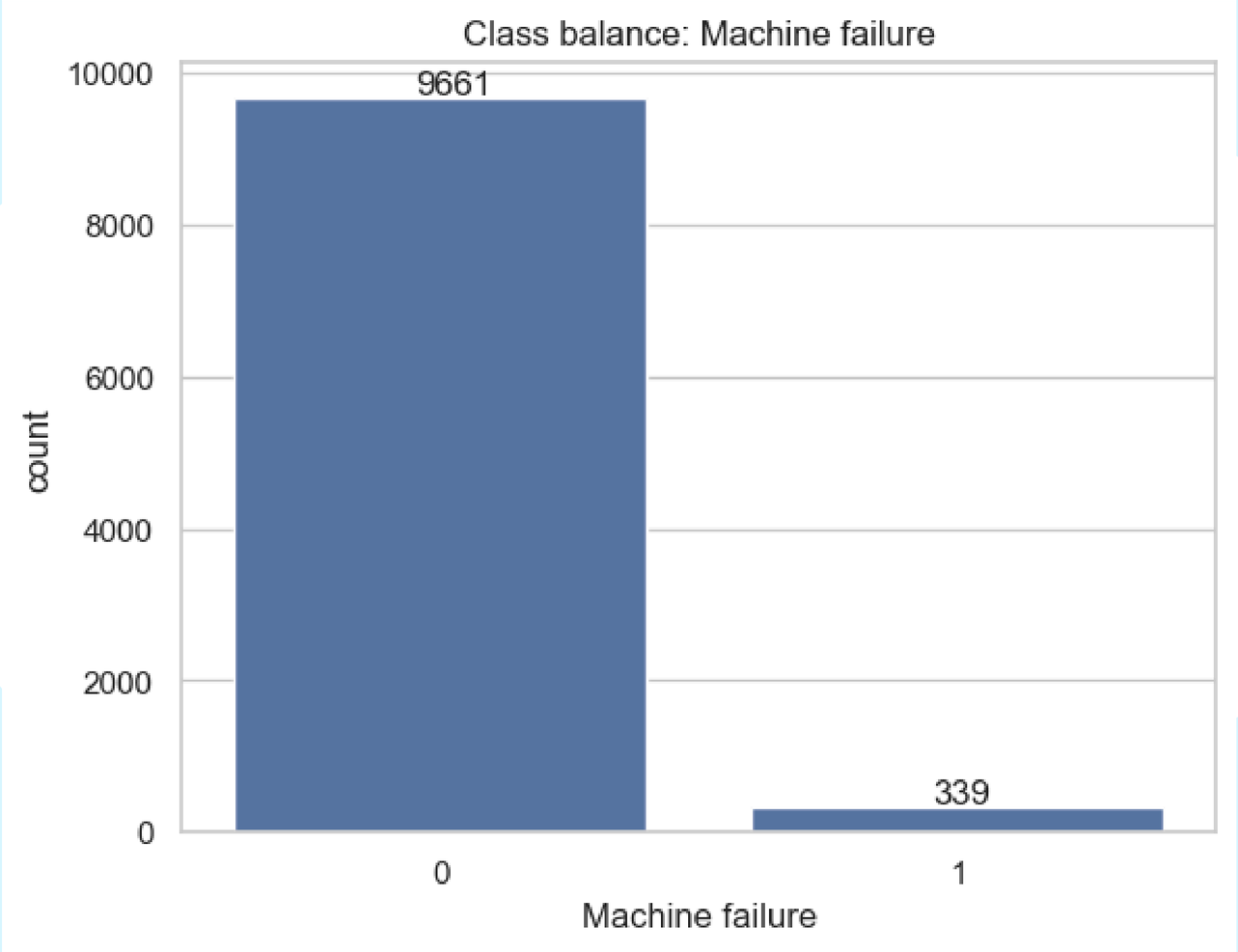
```
Missing values:
 UDI                        0
Product ID                  0
Type                        0
Air temperature [K]         0
Process temperature [K]     0
Rotational speed [rpm]      0
Torque [Nm]                 0
Tool wear [min]             0
Machine failure             0
TWF                         0
HDF                         0
PWF                         0
OSF                         0
RNF                         0
dtype: int64
Duplicate rows: 0
Duplicate UDIs: 0
```

Goal: predict Machine failure (0 or 1)

This dataset is part of the following publication, please cite when using this dataset:
S. Matzka, "Explainable Artificial Intelligence for Predictive Maintenance Applications," 2020 Third International Conference on Artificial Intelligence for Industries (AI4I), 2020, pp. 69-74, doi: 10.1109/AI4I49448.2020.00023.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis



Pre-processing

Pre-processing

แบ่งข้อมูลเพื่อเตรียมไป training model

```
features = ['Type', 'Air temperature [K]',
       'Process temperature [K]', 'Rotational speed [rpm]', 'Torque [Nm]',
       'Tool wear [min]']

x = df[features].copy()
y = df["Machine failure"]
print(x.shape)
print(y.shape)
```

```
# change from str to int64
x["Type"] = x["Type"].map({"L": 0, "M": 1, "H": 2})
```

```
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   UDI                      10000 non-null  int64
 1   Product ID               10000 non-null  object
 2   Type                     10000 non-null  object
 3   Air temperature [K]      10000 non-null  float64
 4   Process temperature [K]  10000 non-null  float64
 5   Rotational speed [rpm]   10000 non-null  int64
 6   Torque [Nm]              10000 non-null  float64
 7   Tool wear [min]          10000 non-null  int64
 8   Machine failure          10000 non-null  int64
 9   TWF                      10000 non-null  int64
 10  HDF                      10000 non-null  int64
 11  PWF                      10000 non-null  int64
 12  OSF                      10000 non-null  int64
 13  RNF                      10000 non-null  int64
dtypes: float64(3), int64(9), object(2)
```

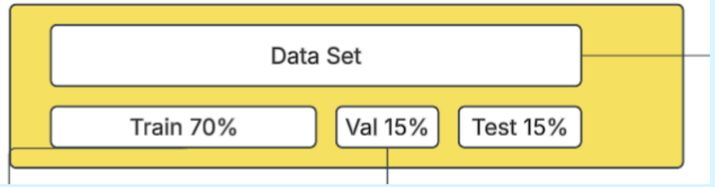
Pre-processing

```python
from sklearn.base import clone
from sklearn.model_selection import GridSearchCV, StratifiedKFold, train_test_split

# Keep the test set untouched until the final evaluation.
x_train, x_temp, y_train, y_temp = train_test_split(
    x,
    y,
    test_size=0.30,
    random_state=RANDOM_STATE,
    stratify=y,
)
x_val, x_test, y_val, y_test = train_test_split(
    x_temp,
    y_temp,
    test_size=0.50,
    random_state=RANDOM_STATE,
    stratify=y_temp,
)

print(f"Training samples:   {len(x_train)}")
print(f"Validation samples: {len(x_val)}")
print(f"Test samples:       {len(x_test)}")
```

```
Training samples:   7000
Validation samples: 1500
Test samples:       1500
```



ทำการดูว่า มีจำนวนค่า 0,1 ใน machine failure เท่าไหร่ ในtraining กับ validation

```python
display(pd.DataFrame({
    "Training": y_train.value_counts(),
    "Validation": y_val.value_counts(),
}).fillna(0).astype(int))
```

	Training	Validation
Machine failure		
0	6763	1449
1	237	51

Modeling and Evaluation

Modeling and Evaluation

+ **No Feature Selection**
+ **Feature Selection: (6)**
  1. ANOVA
  2. Pearson correlation
  3. Lasso (adapt)
  4. Embedded methods
  5. RFE
  6. forward selection and backward selection
+ ***PCA**

+ comparison: uses a fixed random state and class balancing where the estimator supports class_weight="balanced"

The models compared in this notebook are:

1. Support Vector Classifier (SVC)
2. Logistic Regression
3. Decision Tree
4. Random Forest
5. Gradient Boosting Classifier
6. Neural Network MLP

Evaluation Metrics: Because failures are only **3.39%** of the data, accuracy is misleading. We rank models by:

1. **Failure F1-Score**
2. **Failure Recall (missed failures)**
3. **Failure Precision (false alarms)**
4. **Accuracy**

Without Feature Selection

Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
No feature selection	Type, Air temperature [K], Process temperature...	MLP	0.986000	0.841412	0.875000	0.686275	0.769231	16	5
No feature selection	Type, Air temperature [K], Process temperature...	Gradient Boosting	0.984667	0.812345	0.888889	0.627451	0.735632	19	4
No feature selection	Type, Air temperature [K], Process temperature...	Random Forest	0.984667	0.783969	0.966667	0.568627	0.716049	22	1
No feature selection	Type, Air temperature [K], Process temperature...	Decision Tree	0.972667	0.777757	0.604167	0.568627	0.585859	22	19
No feature selection	Type, Air temperature [K], Process temperature...	SVC	0.924667	0.904254	0.296053	0.882353	0.443350	6	107
No feature selection	Type, Air temperature [K], Process temperature...	Logistic Regression	0.830000	0.826879	0.145833	0.823529	0.247788	9	246



Filter: ANOVA filter selection

ANOVA-selected features (4): ['Air temperature [K]', 'Rotational speed [rpm]', 'Torque [Nm]', 'Tool wear [min]']

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
5	ANOVA	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
4	ANOVA	Air temperature [K], Rotational speed [rpm], T...	Gradient Boosting	0.977333	0.723420	0.793103	0.450980	0.575000	28	6
3	ANOVA	Air temperature [K], Rotational speed [rpm], T...	Random Forest	0.972667	0.654792	0.727273	0.313725	0.438356	35	6
0	ANOVA	Air temperature [K], Rotational speed [rpm], T...	SVC	0.918000	0.919722	0.283133	0.921569	0.433180	4	119
2	ANOVA	Air temperature [K], Rotational speed [rpm], T...	Decision Tree	0.961333	0.658385	0.414634	0.333333	0.369565	34	24
1	ANOVA	Air temperature [K], Rotational speed [rpm], T...	Logistic Regression	0.811333	0.788840	0.125806	0.764706	0.216066	12	271



Filter: Pearson correlation selection


```
Pearson-selected features (4): ['Torque [Nm]', 'Tool wear [min]', 'Air temperature [K]', 'Rotational speed [rpm]']
```

	absolute Pearson correlation
Torque [Nm]	0.190886
Tool wear [min]	0.098262
Air temperature [K]	0.092141
Rotational speed [rpm]	0.047755
Process temperature [K]	0.046375
Type	0.038291

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
5	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
4	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	Gradient Boosting	0.977333	0.723420	0.793103	0.450980	0.575000	28	6
3	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	Random Forest	0.972667	0.654792	0.727273	0.313725	0.438356	35	6
0	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	SVC	0.918000	0.919722	0.283133	0.921569	0.433180	4	119
2	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	Decision Tree	0.961333	0.658385	0.414634	0.333333	0.369565	34	24
1	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	Logistic Regression	0.811333	0.788840	0.125806	0.764706	0.216066	12	271

Filter: Adapted Lasso selection


```
Adapted Lasso-selected features (4): ['Torque [Nm]', 'Rotational speed [rpm]', 'Air temperature [K]', 'Tool wear [min]']
```

	absolute L1 coefficient
Torque [Nm]	2.296194
Rotational speed [rpm]	1.663492
Air temperature [K]	0.926099
Tool wear [min]	0.633555
Process temperature [K]	0.483274
Type	0.142156

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
5	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
4	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	Gradient Boosting	0.977333	0.723420	0.793103	0.450980	0.575000	28	6
3	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	Random Forest	0.972667	0.654792	0.727273	0.313725	0.438356	35	6
0	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	SVC	0.918000	0.919722	0.283133	0.921569	0.433180	4	119
2	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	Decision Tree	0.961333	0.658385	0.414634	0.333333	0.369565	34	24
1	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	Logistic Regression	0.811333	0.788840	0.125806	0.764706	0.216066	12	271

Embedded methods

Regularization	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
Ridge (L2)	0.830000	0.826879	0.145833	0.823529	0.247788	9	246
Lasso (L1)	0.830000	0.826879	0.145833	0.823529	0.247788	9	246
Elastic Net (L1 + L2)	0.828667	0.665388	0.097656	0.490196	0.162866	26	231



Recursive Feature Elimination (RFE)

Support Estimators as RFE selectors

	Feature 1	Feature 2	Feature 3	Feature 4
Logistic Regression	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]
Linear SVC	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]
Decision Tree	Air temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]
Random Forest	Air temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]

RFE use Logistic Regression as the selector

```
Active RFE selector (Logistic Regression) features: ['Air temperature [K]', 'Process temperature [K]', 'Rotational speed [rpm]', 'Torque [Nm]']
```

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
5	RFE	Air temperature [K], Process temperature [K], ...	MLP	0.981333	0.820079	0.767442	0.647059	0.702128	18	10
3	RFE	Air temperature [K], Process temperature [K], ...	Random Forest	0.983333	0.773820	0.933333	0.549020	0.691358	23	2
4	RFE	Air temperature [K], Process temperature [K], ...	Gradient Boosting	0.983333	0.764361	0.964286	0.529412	0.683544	24	1
2	RFE	Air temperature [K], Process temperature [K], ...	Decision Tree	0.972000	0.767954	0.595745	0.549020	0.571429	23	19
0	RFE	Air temperature [K], Process temperature [K], ...	SVC	0.898667	0.834044	0.217877	0.764706	0.339130	12	140
1	RFE	Air temperature [K], Process temperature [K], ...	Logistic Regression	0.813333	0.818252	0.134185	0.823529	0.230769	9	271

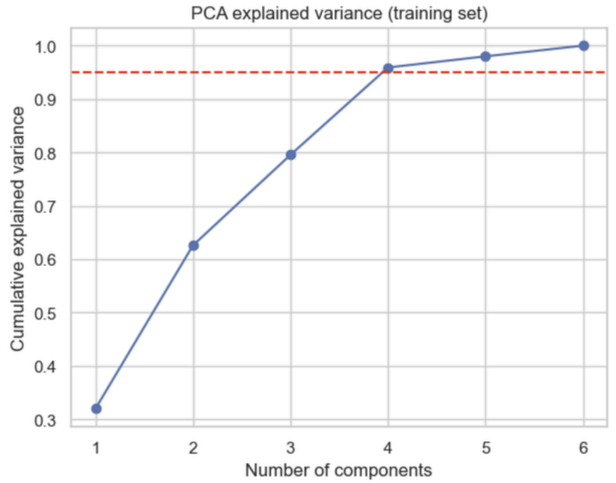
Forward and Backward Selection


```
Running Forward & Backward SFS using estimator: Decision Tree
Forward-selected features:  ['Air temperature [K]', 'Rotational speed [rpm]', 'Torque [Nm]', 'Tool wear [min]']
Backward-selected features: ['Air temperature [K]', 'Process temperature [K]', 'Torque [Nm]', 'Tool wear [min]']
```

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
5	Forward selection	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
4	Forward selection	Air temperature [K], Rotational speed [rpm], T...	Gradient Boosting	0.977333	0.723420	0.793103	0.450980	0.575000	28	6
3	Forward selection	Air temperature [K], Rotational speed [rpm], T...	Random Forest	0.972667	0.654792	0.727273	0.313725	0.438356	35	6
0	Forward selection	Air temperature [K], Rotational speed [rpm], T...	SVC	0.918000	0.919722	0.283133	0.921569	0.433180	4	119
2	Forward selection	Air temperature [K], Rotational speed [rpm], T...	Decision Tree	0.961333	0.658385	0.414634	0.333333	0.369565	34	24
1	Forward selection	Air temperature [K], Rotational speed [rpm], T...	Logistic Regression	0.811333	0.788840	0.125806	0.764706	0.216066	12	271

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
5	Backward selection	Air temperature [K], Process temperature [K], ...	MLP	0.979333	0.762290	0.794118	0.529412	0.635294	24	7
4	Backward selection	Air temperature [K], Process temperature [K], ...	Gradient Boosting	0.974667	0.712581	0.709677	0.431373	0.536585	29	9
0	Backward selection	Air temperature [K], Process temperature [K], ...	SVC	0.900667	0.910750	0.244792	0.921569	0.386831	4	145
2	Backward selection	Air temperature [K], Process temperature [K], ...	Decision Tree	0.959333	0.676268	0.395833	0.372549	0.383838	32	29
3	Backward selection	Air temperature [K], Process temperature [K], ...	Random Forest	0.968000	0.633459	0.560000	0.274510	0.368421	37	11
1	Backward selection	Air temperature [K], Process temperature [K], ...	Logistic Regression	0.758667	0.780498	0.104326	0.803922	0.184685	10	352

PCA - Principal Component Analysis



Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
PCA	4 components	MLP	0.976000	0.675435	0.857143	0.352941	0.500000	33	3
PCA	4 components	Gradient Boosting	0.973333	0.683514	0.703704	0.372549	0.487179	32	8
PCA	4 components	Random Forest	0.970000	0.643953	0.625000	0.294118	0.400000	36	9
PCA	4 components	Decision Tree	0.962667	0.668534	0.439024	0.352941	0.391304	33	23
PCA	4 components	SVC	0.878000	0.861182	0.197248	0.843137	0.319703	8	175
PCA	4 components	Logistic Regression	0.702667	0.751512	0.085954	0.803922	0.155303	10	436

Feature Selection Methods Compared (k = 4)

Category	Method	Features Selected ($k = 4$)	Features Dropped
Baseline	**No feature selection**	All 6 features (`Type` , `Air temp` , `Process temp` , `RPM` , `Torque` , `Tool wear`)	*None*
Filter	**ANOVA & Pearson Correlation**	`Air temperature [K]` , `Rotational speed [rpm]` , `Torque [Nm]` , `Tool wear [min]`	`Type` , `Process temperature [K]`
Filter	**Adapted Lasso (L1)**	`Air temperature [K]` , `Rotational speed [rpm]` , `Torque [Nm]` , `Tool wear [min]`	`Type` , `Process temperature [K]`
Wrapper	**RFE (Logistic Regression)**	`Air temperature [K]` , `Process temperature [K]` , `Rotational speed [rpm]` , `Torque [Nm]`	`Type` , `Tool wear [min]`
Wrapper	**Forward SFS (Decision Tree)**	`Air temperature [K]` , `Rotational speed [rpm]` , `Torque [Nm]` , `Tool wear [min]`	`Type` , `Process temperature [K]`
Wrapper	**Backward SFS (Decision Tree)**	`Air temperature [K]` , `Process temperature [K]` , `Torque [Nm]` , `Tool wear [min]`	`Type` , `Rotational speed [rpm]`
Dim. Reduction	**PCA (`PCA_VARIANCE = 0.95`)**	`4 orthogonal components`	*(Combines all 6 features)*

Compare the experiments

เปรียบเทียบ model ที่ดีที่สุด ของแต่ละการเลือก feature

Best validation model from each feature-selection method:

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
0	No feature selection	Type, Air temperature [K], Process temperature...	MLP	0.986000	0.841412	0.875000	0.686275	0.769231	16	5
1	RFE	Air temperature [K], Process temperature [K], ...	MLP	0.981333	0.820079	0.767442	0.647059	0.702128	18	10
2	ANOVA	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
3	Pearson correlation	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
4	Adapted Lasso	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
5	Forward selection	Air temperature [K], Rotational speed [rpm], T...	MLP	0.980667	0.791357	0.789474	0.588235	0.674157	21	8
6	Backward selection	Air temperature [K], Process temperature [K], ...	MLP	0.979333	0.762290	0.794118	0.529412	0.635294	24	7
7	PCA	4 components	MLP	0.976000	0.675435	0.857143	0.352941	0.500000	33	3



Compare the experiments

	Feature selection	Features	Model	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
0	No feature selection	Type, Air temperature [K], Process temperature...	MLP	0.986000	0.841412	0.875000	0.686275	0.769231	16	5
1	No feature selection	Type, Air temperature [K], Process temperature...	Gradient Boosting	0.984667	0.812345	0.888889	0.627451	0.735632	19	4
2	No feature selection	Type, Air temperature [K], Process temperature...	Random Forest	0.984667	0.783969	0.966667	0.568627	0.716049	22	1
3	No feature selection	Type, Air temperature [K], Process temperature...	Decision Tree	0.972667	0.777757	0.604167	0.568627	0.585859	22	19
4	No feature selection	Type, Air temperature [K], Process temperature...	SVC	0.924667	0.904254	0.296053	0.882353	0.443350	6	107
5	No feature selection	Type, Air temperature [K], Process temperature...	Logistic Regression	0.830000	0.826879	0.145833	0.823529	0.247788	9	246
...										
42	PCA	4 components	MLP	0.976000	0.675435	0.857143	0.352941	0.500000	33	3
43	PCA	4 components	Gradient Boosting	0.973333	0.683514	0.703704	0.372549	0.487179	32	8
44	PCA	4 components	Random Forest	0.970000	0.643953	0.625000	0.294118	0.400000	36	9
45	PCA	4 components	Decision Tree	0.962667	0.668534	0.439024	0.352941	0.391304	33	23
46	PCA	4 components	SVC	0.878000	0.861182	0.197248	0.843137	0.319703	8	175
47	PCA	4 components	Logistic Regression	0.702667	0.751512	0.085954	0.803922	0.155303	10	436

Model Selection

เลือก feature กับ model ที่ให้ผลลัพธ์ดีที่สุด 3 อันดับแรก

```
Top 3 candidates advancing to Hyperparameter Fine-Tuning (UNIQUE_FINALIST_MODELS=False):
```

	Model	Feature selection	Features	Validation accuracy	Validation balanced accuracy	Validation failure precision	Validation failure recall	Validation failure F1	Validation missed failures	Validation false alarms
0	MLP	No feature selection	Type, Air temperature [K], Process temperature...	0.986000	0.841412	0.875000	0.686275	0.769231	16	5
1	Gradient Boosting	No feature selection	Type, Air temperature [K], Process temperature...	0.984667	0.812345	0.888889	0.627451	0.735632	19	4
2	Random Forest	No feature selection	Type, Air temperature [K], Process temperature...	0.984667	0.783969	0.966667	0.568627	0.716049	22	1

Fine-Tuning

Cross-validation tuning results on training data:

	Feature selection	Model	Features	Input features	Best parameters	CV F1 mean	CV F1 std	CV recall mean	CV precision mean	CV balanced accuracy mean	CV train F1 mean
0	No feature selection	Gradient Boosting	Type, Air temperature [K], Process temperature...	[Type, Air temperature [K], Process temperatur...	{'learning_rate': 0.1, 'max_depth': 5, 'n_esti...	0.735597	0.039728	0.692287	0.791109	0.842891	1.000000
1	No feature selection	Random Forest	Type, Air temperature [K], Process temperature...	[Type, Air temperature [K], Process temperatur...	{'class_weight': None, 'max_depth': None, 'min...	0.695148	0.036931	0.578103	0.873527	0.787573	1.000000
2	No feature selection	MLP	Type, Air temperature [K], Process temperature...	[Type, Air temperature [K], Process temperatur...	{'mlpclassifier__alpha': 0.01, 'mlpclassifier_...	0.694023	0.037135	0.632624	0.777869	0.812911	0.808927

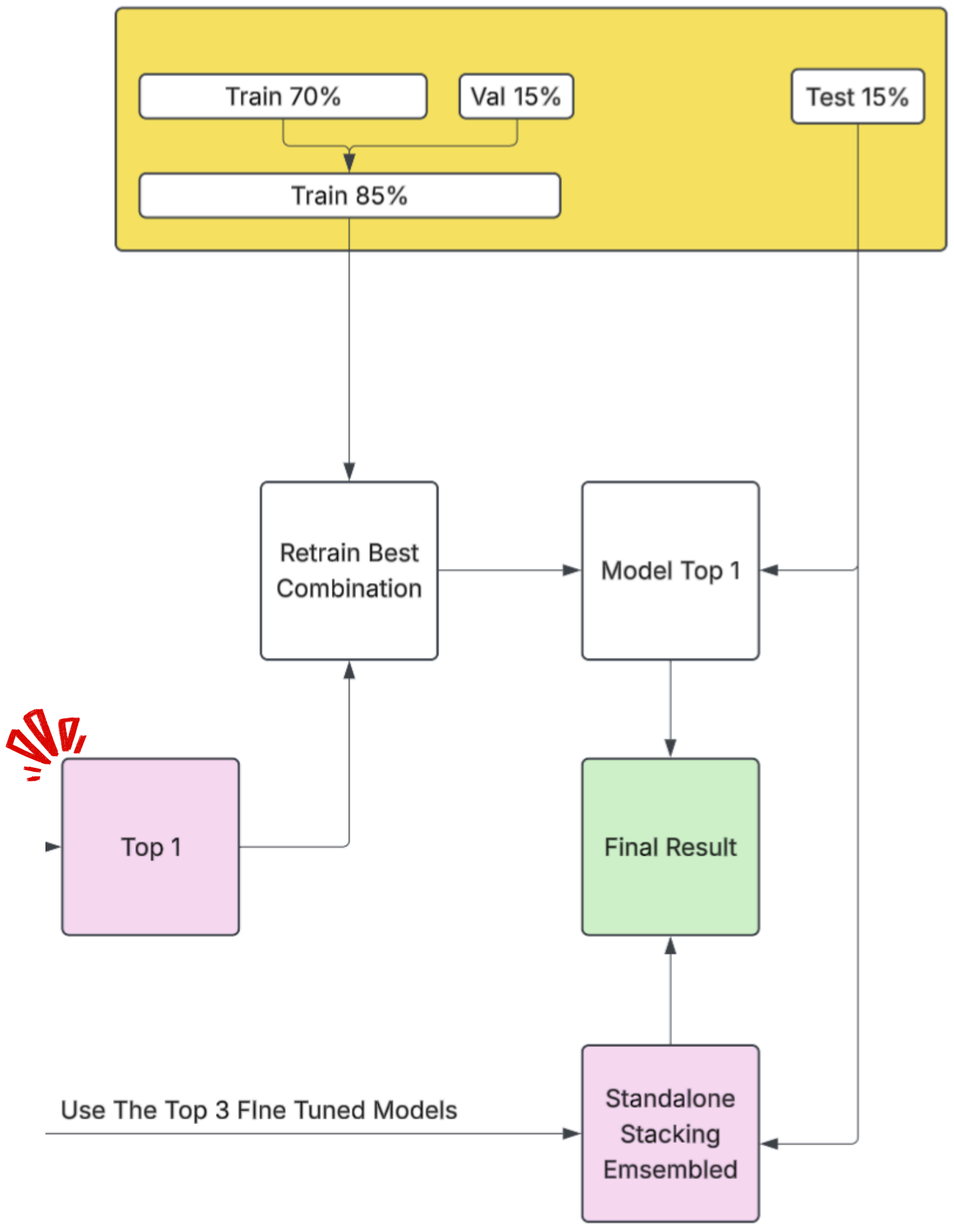
Tuned finalists compared on the Validation split:

	Feature selection	Model	Features	Validation F1	Validation recall	Validation precision	Validation balanced accuracy	Validation missed failures	Validation false alarms
0	No feature selection	Gradient Boosting	Type, Air temperature [K], Process temperature...	0.786517	0.686275	0.921053	0.842102	16	3
1	No feature selection	Random Forest	Type, Air temperature [K], Process temperature...	0.777778	0.686275	0.897436	0.841757	16	4
2	No feature selection	MLP	Type, Air temperature [K], Process temperature...	0.757895	0.705882	0.818182	0.850181	15	8

TOP 1 :

no feature selection

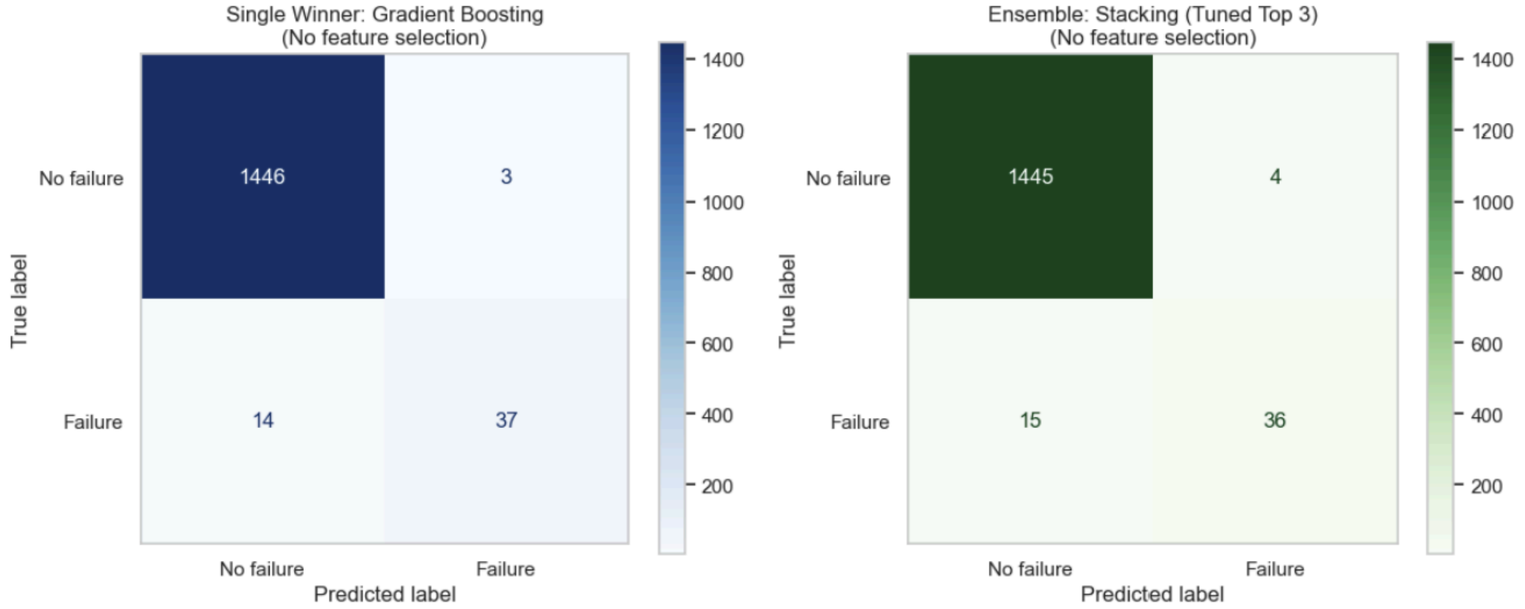
+ Gradient Boosting



Final Model Evaluation

Selected winner: Gradient Boosting (No feature selection)
Selected hyperparameters: {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'subsample': 1.0}

	Feature selection	Model	Features	Test F1	Test recall	Test precision	Test balanced accuracy	Test missed failures	Test false alarms
0	No feature selection	Gradient Boosting	Type, Air temperature [K], Process temperature...	0.813187	0.725490	0.925	0.861710	14	3
1	No feature selection	Stacking (Tuned Top 3)	Type, Air temperature [K], Process temperature...	0.791209	0.705882	0.900	0.851561	15	4

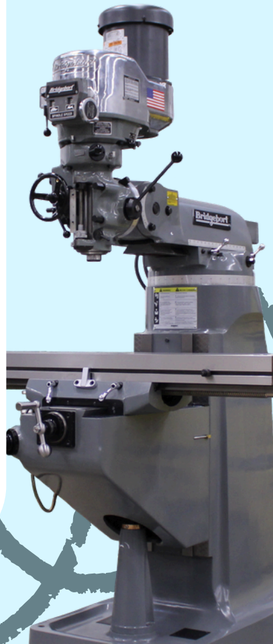


Conclusion

จากการทดลองใช้ feature selection หลายประเภท พบว่า feature ใน datasets ชุดนี้ มีความสำคัญทั้งหมด

ดังนั้น datasets ชุดนี้ ไม่ต้องทำ feature selection เนื่องจากการใช้วิธี no feature selection ให้ค่า f1 score สูงที่สุด

	Feature selection	Model	Features	Test F1	Test recall	Test precision	Test balanced accuracy	Test missed failures	Test false alarms
0	No feature selection	Gradient Boosting	Type, Air temperature [K], Process temperature...	0.813187	0.725490	0.925	0.861710	14	3
1	No feature selection	Stacking (Tuned Top 3)	Type, Air temperature [K], Process temperature...	0.791209	0.705882	0.900	0.851561	15	4



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Thank you